

Notes: Cetorelli & Goldberg (JF, 2012)

Banking Globalization and Monetary Transmission

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1 Big picture

- **Question.** How does banking globalization change the bank-lending channel of monetary policy?
- **Classic lending-channel view.** A monetary tightening reduces bank funding. Banks that cannot replace lost liabilities and lack liquid-asset buffers reduce loan supply.
- **Paper's extension.** For global banks, the liquidity shock does not stop at the domestic balance sheet. The U.S. head office can borrow internally from foreign offices or send less support abroad.
- **Core mechanism.** Global banks manage liquidity at the organization level. Internal capital markets move funds across borders in response to local monetary-policy shocks.
- **Main domestic result.** Large global banks are insulated from U.S. monetary policy shocks, while large domestic banks are not as insulated as the traditional large-bank result suggests.
- **Affiliate result.** Small U.S. banks affiliated with large global bank holding companies are also more insulated than small banks affiliated with large domestic bank holding companies.
- **Direct internal-funding result.** When U.S. monetary policy tightens, global banks increase internal inflows from foreign offices to U.S. parents, measured by changes in Net Due balances.
- **International transmission result.** The same internal capital market transmits U.S. liquidity shocks abroad: foreign-office lending contracts when foreign offices supply more internal funds to constrained U.S. parents.
- **Crisis evidence.** During 2007–2009, weekly balance-sheet data show large movements in cross-border internal borrowing/lending, and low-liquidity banks display a negative relation between internal funding to the parent and foreign lending.
- **Economic takeaway.** Global banking does not weaken monetary transmission; it changes its geography. U.S. monetary policy can affect both domestic lending and foreign lending through the balance sheets of global banks.

2 Data and measurement

2.1 Bank-level data

- The main sample uses quarterly Call Report data for all chartered U.S. banks from 1980:Q1 to 2005:Q4.
- A bank is classified as **global** if it reports positive foreign-office assets.
- A bank is classified as **domestic** if it has no foreign-office assets.
- **Large banks** are banks above the 95th percentile of the asset distribution in each quarter.
- **Small banks** are below the 90th percentile.
- The authors separately analyze small banks affiliated with bank holding companies that contain either large domestic banks or large global banks.

Interpretation: The classification is designed to isolate the organizational feature that matters for the paper: whether the bank can access a cross-border internal capital market.

2.2 Balance-sheet variables

- Total lending and commercial-and-industrial (C&I) lending are the main domestic loan outcomes.
- Liquidity is measured by the log of liquid assets divided by total assets.
- Other controls include capitalization, asset size, nonperforming loans, and local/geographic indicators.

- For global banks, the data also contain foreign-office lending and internal positions with foreign offices.

Interpretation: The liquidity ratio is the balance-sheet buffer. If monetary policy tightens and lending depends more on this ratio, that is interpreted as evidence of a lending channel.

2.3 Foreign lending

- **Foreign loans** are loans booked by the foreign offices of U.S. global banks.
- Foreign loans are distinct from cross-border lending by the U.S. parent.
- Figure 1 shows that global banks are the dominant source of international lending by U.S. banking organizations.

Interpretation: The paper separates international lending by location. This matters because the proposed transmission channel works through foreign offices, not simply through U.S. banks lending cross-border.

2.4 Internal funding: Net Due

- The key internal capital market measure is “Net Due with foreign offices.”
- A positive Net Due To means the U.S. head office owes funds to foreign offices.
- A positive Net Due From means foreign offices owe funds to the U.S. head office.
- In the paper’s constructed variable, a positive change in Net Due indicates an inflow of funds from foreign operations to the U.S. head office.
- Net Due is measured in real 2005Q4 dollars.

Interpretation: Net Due provides unusually direct evidence on internal capital markets. Most papers cannot observe flows inside the firm; here the authors can observe cross-border positions between U.S. parents and foreign offices.

2.5 Monetary policy variables

The authors use three monetary-policy indicators:

- nominal federal funds rate;
- real federal funds rate;
- Bernanke–Mihov liquidity index, modified so that larger values indicate tighter liquidity conditions.

Interpretation: All monetary-policy variables are scaled so that a positive movement means tighter U.S. monetary conditions. This makes coefficient signs comparable across specifications.

3 Empirical specifications

3.1 First-step lending sensitivity regression

For each quarter and each bank group, the authors estimate:

$$\Delta \log(Y_{i,t}) = \sum_{j=1}^4 a_{t,j} \Delta \log(Y_{i,t-j}) + \beta_t X_{i,t-1} + \text{Controls}_{i,t-1} + \varepsilon_{i,t}, \quad (1)$$

where $Y_{i,t}$ is total lending or C&I lending, and $X_{i,t-1}$ is the liquidity-to-asset ratio.

Interpretation: The coefficient β_t measures how dependent lending is on bank liquidity in quarter t . A high β_t means illiquid banks reduce lending more than liquid banks.

3.2 Second-step monetary-policy regression

The time series of first-step coefficients is then regressed on monetary-policy indicators:

$$\beta_t = \eta + \sum_{j=1}^n \phi_j \text{MP}_{t-j} + \delta' \text{Controls}_t + \mu_t, \quad n = 8. \quad (2)$$

Interpretation: A positive and significant $\sum_j \phi_j$ means lending becomes more liquidity-sensitive when monetary policy tightens. This is evidence of an active bank-lending channel.

3.3 Instrumenting liquidity

The liquidity-to-asset ratio may respond to macro conditions. To address this, the paper instruments liquidity using residuals from a regression of liquidity on:

- the ratio of C&I loans to total loans;
- the ratio of nonperforming loans to total loans.

Interpretation: The goal is to purge cyclical loan-composition and credit-quality components from the liquidity measure.

3.4 Internal capital market regression

For global banks, the direct test of internal funding is:

$$\Delta \text{NetDue}_{i,t} = \alpha + \sum_{j=1}^4 \varphi_j \Delta \text{NetDue}_{i,t-j} + \sum_{j=0}^4 \theta_j \Delta \text{MP}_{t-j} + \sum_{j=0}^4 \gamma_j \Delta \text{GDP}_{t-j} + \mu_{i,t}. \quad (3)$$

Interpretation: A positive $\sum_j \theta_j$ means that when U.S. monetary policy tightens, U.S. parent banks receive more internal funding from foreign offices. This is the direct internal-capital-market channel.

3.5 Foreign-office lending dependence on parent liquidity

The authors then repeat the two-step lending-channel design with lending by foreign offices as the dependent variable and the U.S. parent liquidity ratio as the key first-stage variable.

Interpretation: If foreign-office lending becomes less dependent on parent liquidity when U.S. policy tightens, this means foreign offices are supplying funds to parents and relying less on parent balance-sheet support.

3.6 Direct transmission through Net Due

The direct foreign-lending transmission regression is:

$$\begin{aligned} \Delta Y_{i,t} = & \alpha + \sum_{j=1}^4 \Delta \text{NetDue}_{i,t-j} [\beta_j + \gamma_j \Delta \text{MP}_{t-j} + \delta_j \text{LowLiquidity}_{i,t-j} + \eta_j \Delta \text{MP}_{t-j} \times \text{LowLiquidity}_{i,t-j}] \\ & + \text{Controls}_{i,t} + \varepsilon_{i,t}. \end{aligned} \quad (4)$$

Interpretation: The central term is the interaction of Net Due, monetary policy, and low liquidity. A negative effect for low-liquidity banks means that foreign offices reduce external lending when they transfer funds internally to constrained U.S. parents.

4 Identification and empirical design

4.1 What identifies the lending-channel difference

- The baseline compares the sensitivity of lending to liquidity across large global and large domestic banks.
- The comparison is performed using the same two-step methodology as Kashyap and Stein (2000).
- If size alone insulates banks, both large global and large domestic banks should be insulated.
- If globalness matters, large global banks should be more insulated because they can use internal funding from foreign offices.

Interpretation: The key variation is organizational: large banks with global operations versus large banks without global operations.

4.2 Why small affiliates are useful

- Global banks may lend to different customers than domestic banks.
- To reduce loan-demand heterogeneity, the paper studies small U.S. banks affiliated with large domestic or large global bank holding companies.
- Small banks serve more similar local borrowers.
- If small affiliates of global BHCs are insulated while small affiliates of domestic BHCs are not, the result is more likely due to the parent organization's funding network.

Interpretation: The small-affiliate exercise helps separate supply-side funding effects from demand differences in the customer base of large global banks.

4.3 Why Net Due is important

- The lending-channel tests are indirect: they infer funding shocks from lending sensitivity.
- Net Due provides direct evidence of internal capital market movements.
- A systematic Net Due response to monetary policy makes the proposed mechanism observable.

Interpretation: The paper is unusually strong because it does not stop at showing that global banks are insulated. It observes the internal funding channel that creates insulation.

4.4 Why crisis evidence is complementary

- The main regressions end in 2005 to keep monetary policy separate from the 2007–2009 crisis.
- The crisis provides a separate stress episode not directly driven by ordinary monetary-policy shocks.
- Weekly aggregate data show how internal funding behaved when liquidity stress became acute.
- The bank-level crisis scatter plots test whether the same substitution logic holds in a different environment.

Interpretation: The crisis evidence is not the main identification strategy, but it validates the mechanism during a large real-world liquidity shock.

4.5 What the paper cannot fully prove

- Global and domestic banks may still differ along unobserved dimensions.

- Net Due records positions, not every underlying transaction or purpose.
- The analysis focuses on U.S. banks and may not apply one-for-one to other regulatory systems.
- The crisis evidence is descriptive and based on a short period.
- The paper studies loan supply responses but cannot perfectly separate all loan-demand movements.

Interpretation: The empirical case is persuasive because several independent patterns line up: domestic insulation, small-affiliate insulation, direct internal funding movements, foreign lending effects, and crisis-period evidence.

5 Tables and figures

This section preserves the original tables and figures by directly extracting them from the paper. Adjacent visuals are grouped to save space and improve comparison speed.

5.1 Table I + Table II: bank groups, foreign loans, and Net Due

Table I
Basic Balance Sheet Information for U.S. Banks

Data are from quarterly Call Report forms for all banks from 1980Q1 to 2005Q4. A bank is defined as global in a quarter if it reports positive foreign office assets. A bank is defined as domestic if it has no foreign office assets. Large banks are those with total assets above the 95th percentile of the total asset distribution in each quarter. Small banks are those with total assets below the 90th percentile of the total asset distribution in each quarter. Small banks in domestic BHCs are small banks in BHCs with at least one large domestic bank and no global banks. Small banks in global BHCs are small banks in those BHCs with at least one large global bank.

	All Banks	Large Domestic Banks	Large Global Banks	Small in Banks BHCs	Small in Banks Global BHCs
Total number of bank observations (1980Q1–2005Q4)	1,162,969	43,921	14,252	41,339	47,640
Median values for bank asset size (millions 2005USD)					
1985	62.3	997.0	5,123.7	93.9	103.0
1995	73.9	1,775.9	10,358.6	142.7	134.8
2005	105.2	2,236.5	22,300.0	213.3	213.2
Share of each bank group in total assets (%)					
1985	100.0	16.6	56.0	1.4	2.2
1995	100.0	22.6	56.1	1.0	0.9
2005	100.0	17.9	67.9	0.4	0.3
Median total loans/assets (%)	55.6	61.1	60.4	57.1	55.5
Median C&I loans/assets (%)	17.3	22.8	35.4	18.4	21.0
Median real estate loans/assets (%)	24.5	26.0	17.5	26.2	22.3
Median bank liquid assets/total assets (%)	24.1	19.8	16.6	27.3	26.3
Median capitalization ratio (%)	8.7	7.2	6.4	8.0	7.6
Value of nonperforming loans/total loans (%)	1.0	1.0	1.1	1.6	0.8

Table I: balance-sheet characteristics by bank group

Table II
Net Due and Foreign Loans
(Millions 2005 USD)

Net due *to/from* indicate the position of the domestic offices of a bank relative to all of the bank's Edge and Agreement subsidiaries, foreign branches, consolidated foreign subsidiaries, and branches in Puerto Rico and U.S. territories and possessions (schedule RC-H from form FFIEC 031—Call Report). A positive net due *to* indicates that the head office owes funds to its foreign offices. A positive net due *from* indicates that the head office is owed funds from its foreign offices. Foreign loans are the total loans booked by the foreign offices of U.S. global banks.

		1985Q4	1995Q4	2005Q4
Number of global banks		247	170	107
Loans of foreign offices				
Total loans	Median value across banks	\$19.3	\$0.027	\$0
	Mean value across banks	\$1,599.7	\$1,978	\$3,129.8
	Share of total bank lending	0.15	0.11	0.07
C&I Loans	Median value across banks	\$4.8	\$0	\$0
	Mean value across banks	\$866.4	\$942.2	\$1,236.9
	Share of total C&I lending	0.08	0.05	0.03
Net due				
Net due to	Median	\$62.3	\$299.2	\$657.3
	Mean	\$304.3	\$955.7	\$3,856.1
	Number of observations	60	103	62
Net due from	Median	\$43.3	\$3.9	\$852
	Mean	\$458.3	\$332.5	\$984
	Number of observations	187	67	45
(Net due to – net due from)				
	Median absolute value	\$47.3	\$141.9	\$74.4
	Mean absolute value	\$420.9	\$710.1	\$2,648.2
	Number of observations	247	170	107

Table II: foreign loans and Net Due positions

Read:

- Table I shows that large global banks are much larger than large domestic banks and hold a rising share of total banking assets.
- In 2005, large global banks account for about 67.9% of total U.S. banking assets, while large domestic banks account for about 17.9%.
- Large global banks have lower median liquidity-to-asset ratios and lower capitalization ratios than large domestic banks.
- Table II shows that internal positions with foreign offices are large and can run in either direction: some banks report Net Due To and others Net Due From.
- Mean absolute Net Due rises substantially over time, reaching about \$2.65 billion by 2005Q4.

Economic message: Global banks are not simply large domestic banks. They have cross-border balance sheets and measurable internal funding positions, which makes the internal-capital-market hypothesis testable.

5.2 Figure 1 + Table III: international lending and the large-bank lending channel

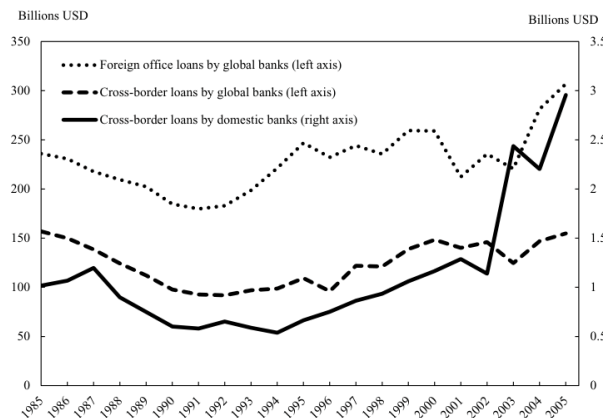


Figure 1. International lending by U.S. banks. Source: Data on loans are from schedule RC-C of form FFIEC 031 of the Call Reports. Cross-border loans are computed as the sum of consolidated C&I loans to non-U.S. addressees (RCFD 1764), domestic offices' loans to banks in foreign countries (RCON B535), consolidated loans to foreign governments and official institutions (including foreign central banks) (RCFD 2081), and consolidated lease financing receivables to non-U.S. addressees (RCFD 2183), sorted every quarter and averaged annually. Foreign loans are computed as the difference between total loans and leases of consolidated banks (RCFD 2122) and total loans of their domestic offices (RCON 2122), sorted every quarter and averaged annually. A bank is defined as global in a quarter if it reports positive foreign assets. A bank is defined as domestic if all its activity comes from offices located domestically.

Figure 1: international lending by U.S. banks

Table III
Lending Channel for Large Domestic and Large Global Banks

Results are from equation (2) using Instrumental Variables specifications for total bank loans. The dependent variable is the time series of estimated coefficients of the liquidity-to-asset ratio from quarterly cross-sectional regressions based on specification (1). The liquidity-to-asset ratio is instrumented by its residuals from a regression on the C&I-to-total lending ratio and the ratio of nonperforming to total loans. The cells present the sum of the estimated coefficients on the eight lags of each respective monetary policy variable. The Bernanke-Mihov index has been modified from the original so that all three monetary policy indicators signal tightening when they increase. Reported in brackets is the probability that the sum of the coefficients is significantly different from zero. The first two columns report results for the group of large domestic banks. The last three columns report results for the group of large global banks. Columns 1 and 3 refer to second-stage specifications without GDP controls, while columns 2, 4, and 5 refer to specifications including GDP controls. Column 5 also includes foreign rate controls. In the lower panel, the first two columns report results for large domestic banks, excluding banks in the top 1% in asset size. Columns 3 and 4 report equivalent results for large global banks. The fifth column report results for large global banks based on a Weighted Least Squares regression, with weights determined using the size distribution of large domestic banks. * indicates statistical significance at the 10% level or better. Sample period: 1980:Q1–2005:Q4. Standard errors are computed with an eight-lag Newey–West (1987) correction for autocorrelation and heteroskedasticity.

	Large Domestic Banks		Large Global Banks		
	No GDP controls (1)	With GDP controls (2)	No GDP controls (3)	With GDP controls (4)	GDP and foreign rate controls (5)
Federal funds rate (nominal)	0.0008* [0.008]	0.0008* [0.006]	−0.0001 [0.915]	−0.0004 [0.659]	0.0007 [0.637]
Federal funds rate (real)	0.0004 [0.274]	0.0008* [0.036]	0.0017 [0.109]	0.0011 [0.296]	0.0010 [0.519]
Bernanke–Mihov index (negative*100)	0.0004* [0.038]	0.0003* [0.084]	−0.0001 [0.902]	−0.0001 [0.831]	0.0001 [0.820]
Robustness					
	Large Domestic banks		Large Global banks		WLS
	No top 1% No GDP controls (6)	With GDP controls (7)	No top 1% No GDP controls (8)	With GDP controls (9)	
Federal funds rate (nominal)	0.0016* [0.000]	0.0016* [0.000]	−0.0030 [0.444]	−0.0029 [0.509]	0.0026 [0.587]
Federal funds rate (real)	0.0010* [0.077]	0.0014* [0.007]	−0.0012 [0.806]	0.0002 [0.963]	0.0057 [0.296]
Bernanke–Mihov index (negative*100)	0.0005* [0.052]	0.0004 [0.130]	−0.0032 [0.264]	−0.0032 [0.334]	−0.0018 [0.589]

Table III: lending-channel tests for large banks

Read:

- Figure 1 shows that foreign-office loans by global banks and cross-border loans by global banks are much larger than cross-border loans by domestic banks.
- Table III shows that large domestic banks have positive and significant lending-channel coefficients in many specifications.
- Large global banks generally do not have significant positive coefficients.
- Excluding the top 1% of banks and using weighted least squares does not restore a significant lending-channel effect for global banks.

Economic message: Global operations appear to insulate lending from domestic monetary-policy shocks. Size alone is not enough to explain the insulation.

5.3 Table IV + Table V: small affiliates and internal funding

Table IV
Results for Small Affiliated with Large Domestic or Globally Oriented Banks

Specifications are similar to those of Table III, but based on Campello (2002): the objects of analysis are small banks (asset size below the 90th percentile) affiliated with either large domestic banks or large global banks as part of the same BHC. The dependent variable is the time series of estimated coefficients on the net income-to-loan ratio in quarterly cross-sectional regressions where the dependent variable is either growth in total bank loans or total C&I loans. The reported figures in the columns are from the sum of the estimated coefficients on the eight lags of each respective monetary policy variable. The Bernanke–Mihov index has been modified from the original so that all three monetary policy indicators signal tightening when they increase. Reported in brackets are the probabilities that the sum of the coefficients is significantly different from zero. The upper panel reports results from estimations where the dependent variable in the first-step regressions is total lending growth. The lower panel reports results from estimations where the dependent variable in the first-step regressions is total C&I lending growth. The first two columns reports results for the group of small bank members of BHCs where there is at least one large domestic bank and no global banks. The last two columns report results for the group of small bank members of BHCs where there is at least one large global bank and no other large domestic bank. Odd columns refer to second-step specifications without GDP controls, while even columns refer to specifications including GDP controls. * indicates statistical significance at the 10% level or better. Due to sample size constraints, the sample period is 1980:Q1–1996:Q4. Standard errors are computed with an eight-lag Newey–West (1987) correction for autocorrelation and heteroskedasticity.

	Total Bank Lending			
	Small in domestic BHC		Small in global BHC	
	No GDP controls	With GDP controls	No GDP controls	With GDP controls
	(1)	(2)	(3)	(4)
Federal funds rate (nominal)	0.2903* [0.047]	0.5203* [0.026]	−1.1976 [0.131]	−0.3404 [0.471]
Fed funds rate (real)	0.8440* [0.000]	0.9411* [0.001]	−1.5803* [0.057]	−1.8704 [0.142]
Bernanke–Mihov index (negative*100)	0.1278 [0.122]	0.2495* [0.016]	−0.6966* [0.075]	−0.4937* [0.042]
	Total C&I Lending			
	Small in domestic BHC		Small in global BHC	
	No GDP Controls	With GDP Controls	No GDP Controls	With GDP Controls
	(5)	(6)	(7)	(8)
Federal funds rate (nominal)	1.4342* [0.029]	1.0752* [0.087]	−0.1390 [0.764]	−0.0207 [0.970]
Fed funds rate (real)	2.5029* [0.050]	2.6469* [0.027]	−1.0854 [0.233]	−1.0579 [0.282]
Bernanke–Mihov index (negative*100)	0.7712* [0.035]	0.6619* [0.057]	−0.1084 [0.604]	−0.0145 [0.953]

Table IV: small banks affiliated with domestic/global BHCs

Table V
Internal Lending between Parent Banks and Foreign Affiliates

Results are from regressions where the dependent variable is the quarterly real change in net due flows from foreign affiliates to the head office. A positive change indicates a net inflow of funds from foreign operations. The dependent variable is in real 2005Q4 dollars. The reported figures in the columns are from the sum of the estimated coefficients on the eight lags of each respective monetary policy variable. The Bernanke–Mihov index has been modified from the original so that all three monetary policy indicators signal tightening when they increase. In the upper panel, column 1 reports results from the baseline specification of equation (3). Column 2 reports results of the baseline specification where foreign rate controls are also included. Columns 3 and 4 split the sample into periods of monetary tightening and monetary expansion, respectively. Tests of the equality of each pair of estimates from columns 3 and 4 were run but are reported in the Internet Appendix. In the lower panel, columns 1 and 2 split the sample between observations with a capital-to-asset ratio below and above the median, respectively. Columns 3 and 4 instead split the sample into observations below and above the median in asset size. Reported in brackets are the probabilities that the sum of the coefficients is significantly different from zero. * indicates statistical significance at the 10% level or better. Sample period: 1980:Q1–2005:Q4. Regression results have robust standard errors and residuals clustered by bank identifier.

Monetary Variable	Baseline (1)	Baseline with Foreign Rate Controls (2)	Baseline with Potential Asymmetry of Effects	
			Tighter Money (3)	Looser Money (4)
Federal funds rate (nominal)	74,268* [0.026]	80,162* [0.020]	131,158* [0.043]	82,441* [0.000]
Fed funds rate (real)	75,715* [0.044]	104,688* [0.010]	82,266* [0.024]	164,481* [0.020]
Bernanke–Mihov index (negative*100)	14,633* [0.083]	17,918* [0.010]	23,969 [0.230]	24,231* [0.043]
Monetary Variable	Capitalization Ratio		Bank Size	
	Lower (5)	Higher (6)	Below median (7)	Above median (8)
Federal funds rate (nominal)	157,352* [0.008]	−15,562 [0.489]	1,882 [0.698]	152,704* [0.023]
Fed funds rate (real)	163,302* [0.017]	−20,615 [0.288]	−2,025 [0.634]	152,473* [0.043]
Bernanke–Mihov index (negative*100)	28,300* [0.078]	2,598 [0.787]	3,039 [0.362]	30,605* [0.085]

Table V: internal lending between parents and foreign affiliates

Read:

- Table IV shows that small banks affiliated with large domestic BHCs have strong positive lending-channel coefficients.
- Small banks affiliated with large global BHCs are insulated; coefficients are often insignificant or negative.
- Table V provides direct evidence of internal capital markets: monetary tightening is associated with increased Net Due inflows to the U.S. parent.
- The internal funding response is stronger for lower-capitalization banks and larger global banks.
- The response is generally symmetric across tighter and looser monetary-policy periods.

Economic message: The insulation of global banks is not just a demand-side artifact. Internal funding moves across the global banking organization in response to U.S. liquidity conditions.

5.4 Table VI + Table VII: foreign lending and international transmission

Table VI
Monetary Policy and Foreign Lending

Results are based on specifications similar to those reported in Table III. Here the focus is on the activity of the foreign offices of global banks. The dependent variable is the time series of estimated coefficients of the liquidity-to-asset ratio from quarterly cross-sectional regressions where the dependent variable is either growth in total loans or total C&I loans of the foreign offices of global banks. The reported figures in the columns are from the sum of the estimated coefficients on the eight lags of each respective monetary policy variable. The Bernanke-Mihov index has been modified from the original so that all three monetary policy indicators signal tightening when they increase. Reported in brackets are the probabilities that the sum of the coefficients is significantly different from zero. The first two columns report results from estimations where the dependent variable in the first-step regressions is total C&I lending growth of foreign offices. Odd columns refer to second-step specifications without GDP controls, while even columns refer to specifications including GDP controls. All reported specifications instrument the liquidity-to-asset ratio with the series of the residuals of a regression of such variable on the C&I-to-total lending ratio and the ratio of nonperforming to total loans. * indicates statistical significance at the 10% level or better. Sample period: 1980:Q1–2005:Q4. Standard errors are computed with an eight-lag Newey–West (1987) correction for autocorrelation and heteroskedasticity.

Monetary Variable	Total Foreign Lending		Total Foreign C&I Lending	
	Without GDP controls (1)	With GDP controls (2)	Without GDP controls (3)	With GDP controls (4)
Federal funds rate (nominal)	–0.0104* [0.008]	–0.0118* [0.001]	–0.0159* [0.004]	–0.0146* [0.011]
Fed funds rate (real)	–0.0098* [0.049]	–0.0124* [0.002]	–0.0114* [0.021]	–0.0118* [0.025]
Bernanke–Mihov index (negative*100)	–0.0057* [0.045]	–0.0063* [0.016]	–0.0026 [0.469]	–0.0012 [0.768]

Table VI: monetary policy and foreign lending

Table VII
Net Due Effects on Foreign Lending in Response to Monetary Policy

This table presents results from regressions of equation (4). They capture the direct effect of a change in net due in response to a change in monetary policy on foreign offices' lending. We report only a partial list of coefficients, to focus on the effect on low liquidity banks. The dependent variable is either the quarterly change in foreign total lending or foreign C&I lending. Each column is a separate regression for each of the three measures of monetary policy. The reported numbers are from each respective sum of estimated coefficients, as indicated by each row legend. Net due is in real 2005Q4 dollars. Low liquidity is a dummy equal to one if a global bank has a value of liquidity-to-asset ratio below the median in a quarter. Reported in brackets are the probabilities that the sum of the coefficients is significantly different from zero. * indicates statistical significance at the 10% level or better. Sample period: 1980:Q1–2005:Q4.

Regression coefficients	Total Foreign Lending		
	Federal funds rate (nominal)	Fed funds rate (real)	Bernanke–Mihov index (negative*100)
Net due	–0.0232 [0.168]	0.0058 [0.734]	–0.0533* [0.002]
Net due × monetary policy	0.2984* [0.000]	0.2711* [0.000]	0.1243* [0.000]
Net due × low liquidity	–0.0854* [0.001]	–0.1165* [0.000]	–0.0520* [0.044]
Net due × mon policy × low liquidity	–0.2781* [0.000]	–0.2156* [0.000]	–0.1137* [0.000]

	Total Foreign C&I Lending		
	Federal funds rate (nominal)	Fed fundsrate (real)	Bernanke–Mihov index (negative*100)
Net due	0.0097 [0.241]	0.0166* [0.047]	–0.0179* [0.037]
Net due × monetary policy	0.2161* [0.000]	0.1436* [0.000]	0.0969* [0.000]
Net due × low liquidity	–0.0352* [0.015]	–0.0423* [0.003]	–0.0080 [0.574]
Net due × mon policy × low liquidity	–0.1866* [0.000]	–0.1022* [0.000]	–0.0734* [0.000]

Table VII: Net Due effects on foreign lending

Read:

- Table VI shows that foreign lending by U.S. bank affiliates abroad relies less on parent balance-sheet strength when U.S. monetary policy tightens.
- Coefficients are negative and significant in most total foreign lending specifications.
- Table VII shows the direct link between internal transfers and foreign lending.
- The interaction of Net Due, monetary policy, and low liquidity is negative and significant across specifications.
- The paper translates these estimates into an elasticity: for each dollar of extra internal lending received by liquidity-constrained parents, foreign external lending falls by about 29 cents.

Economic message: The internal capital market that protects the U.S. parent transmits the shock abroad. Domestic monetary policy therefore affects foreign credit supply through global banks.

5.5 Figure 2 + Figure 3: crisis-period internal funding and foreign lending

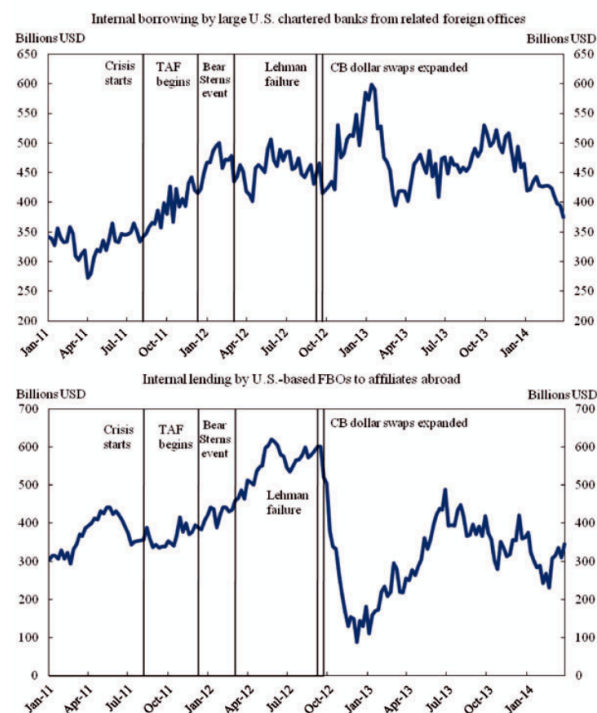


Figure 2. Patterns of internal borrowing during the crisis. Source: Weekly series on aggregate Assets and Liabilities of Commercial Banks in the United States (H.8 report). Data are reported from the perspective of the reporting institution located in the United States. Hence, the series shown are *internal borrowing* by large U.S. chartered commercial banks (net due to related foreign offices), and *internal lending* of U.S.-based foreign banking organizations to their foreign affiliates (negative net due to foreign affiliates).

Figure 2: internal borrowing/lending during the crisis

Read:

- Figure 2 shows a sharp rise in internal borrowing by large U.S. chartered banks from related foreign offices during the crisis.
- The lower panel shows a distinct pattern for U.S.-based foreign banking organizations, especially around the TAF and central bank dollar swap expansion.
- Figure 3 shows that low-liquidity banks have a negative relationship between the pre-post change in Net Due and the pre-post change in foreign lending.
- High-liquidity banks do not show the same negative relationship.

Economic message: The crisis evidence reinforces the regression results: when balance sheets are constrained, internal support to the parent comes at the expense of foreign external lending.

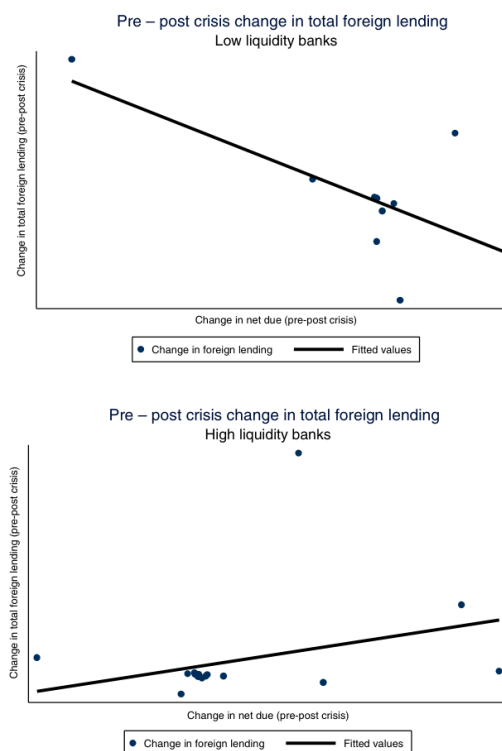


Figure 3. Foreign lending during the crisis. The charts display the relationship between the change in total foreign lending and the change in net due from pre-crisis levels (average levels in quarters up to 2007Q2 compared to average levels afterwards) separately for large global banks with pre-crisis liquid asset levels below median (low liquidity banks) and above median (high liquidity banks). Low liquidity banks display a negative relationship, consistent with the hypothesis that an increase in internal lending is associated with a decrease in external lending

Figure 3: foreign lending during the crisis

6 Short summary

- Cetorelli and Goldberg study how global banking changes monetary transmission.
- The traditional bank-lending channel focuses on domestic banks' inability to replace lost funding.
- Global banks can activate cross-border internal capital markets.
- Large global banks are insulated from U.S. monetary-policy shocks.
- Large domestic banks and small affiliates of domestic BHCs remain sensitive to monetary policy.
- Small banks affiliated with global BHCs are more insulated.
- Net Due data directly show internal transfers between U.S. parents and foreign offices.
- When U.S. monetary policy tightens, U.S. parent banks receive more funds from foreign offices.
- Foreign-office lending contracts when foreign offices provide internal support to liquidity-constrained parents.
- During the financial crisis, internal funding flows moved sharply, consistent with the paper's mechanism.
- The lending channel is therefore larger and more international than domestic-only evidence suggests.

7 Conclusion

7.1 Core conclusion

The paper's central conclusion is that banking globalization changes the scope of monetary transmission. Global banks can shield domestic lending from local monetary shocks by using internal capital markets with foreign offices. But this insulation does not make the shock disappear. It reallocates the shock across borders and can reduce lending abroad.

7.2 Economic intuition

A global bank is a network of balance sheets. When U.S. liquidity tightens, the head office can draw resources from its foreign offices or reduce support to them. This keeps the U.S. parent from cutting domestic lending as much as a nonglobal bank would. However, the foreign offices then have fewer resources for external lending, especially when the global organization is liquidity constrained.

7.3 Takeaway

Global banks transform the bank-lending channel from a domestic mechanism into an international propagation mechanism. Monetary policy works not only through local bank balance sheets, but also through the internal capital markets of globally active banking organizations.